

KisanAlert

Smart Crop Advisory. Any Language. Any Phone.

Track 4: Kisan Alert - Smart Water, Crop & Advisory System

Build with AI: Code for Communities | Google Cloud x Hack2Skill

Gemini 1.5 Flash

Multilingual

SMS + WhatsApp

RSK Integration

The Problem No App Has Solved

86% of Indian farmers are small/marginal - and completely invisible to agri-tech.

2 ha

Average farm size. No soil test. No forecast.
No advisory.

1:800

Extension worker to farmer ratio. Doctor
ratio is 1:1000.

Oct

When a Kharif farmer in Warangal finds out
his crop has failed.

0

Existing solutions that work on a Rs.800
feature phone, in Telugu.

"Farmers make crop decisions based on habit or hearsay, not soil health or groundwater data."

- MP problem statement, Track 4

Three Things. Integrated. In the Farmer's Language.

01

Crop Recommendation

Real soil data + groundwater depth + rainfall forecast ranked by Gemini.

Not guesswork. A scored list in Telugu, Hindi, or Kannada.

02

Dry Spell Alerts

Proactive SMS/voice call 5-7 days before a dry spell hits.

Not reactive. Not after the crop is already stressed.

03

Disease Diagnosis

Farmer sends a photo. Gemini Vision returns disease, severity, treatment.

If confidence < 72%, RSK officer is auto-assigned within 2 hours.

All three talk to each other. One farmer. One conversation. All three working.

Works on 2G SMS. No app install. No smartphone required.

How It Works

Gemini at the centre. Existing infra at the edges.



Layer	Technology	Purpose
Intelligence / NLU	Gemini 1.5 Flash	Understand farmer intent in 5 Indic languages
Crop Disease Vision	Gemini 1.5 Flash Vision	Diagnose disease from crop photo
Voice I/O	Google Cloud STT + TTS	Spoken queries + voice responses
Backend / API	FastAPI on Python 3.11	Stateless, containerised
Deployment	Google Cloud Run	Auto-scales, pay-per-request
Soil + Weather Data	SoilGrids - ISRO Bhuvan - Open-Meteo - IMD	Real satellite + forecast data

The AI Is Doing Real Work

Not a chatbot with canned answers. Let's be specific.

Crop Recommendation

Soil pH 6.4, sandy loam, GW 8m, 742mm kharif rainfall
+ Open-Meteo 7-day forecast
= Gemini ranks Cotton #1, Tur Dal #2, Groundnut #3
With Telugu reasoning. Not a rule lookup.

Disease Diagnosis

Farmer photos yellowing groundnut leaf in Guntur.
Gemini Vision: Alternaria leaf spot, medium severity.
Output: Mancozeb 2g/L, organic fallback, RSK flag if
confidence < 72%.
Structured JSON output, tuned prompt.

Multilingual Intent

"meri fasal mein kuch kharaabi lag rahi hai"
Gemini detects advisory intent > routes to disease
module
> asks for photo in Hindi.
No keyword matching. Actual NLU.

SMS & Voice First - Built for Actual Farmers

Because a smartphone app would reach the wrong farmers.

- 1 Farmer calls KisanAlert number or SMS short code
- 2 IVR: "Hindi mein baat karein / Telugu lo matladandi"
- 3 Farmer speaks or types query in their language
- 4 Cloud STT converts to text. Gemini processes intent.
- 5 Response via TTS voice call OR plain SMS
- 6 WhatsApp photo -> disease diagnosis -> RSK escalation

Languages Supported

Telugu

Hindi

Kannada

Marathi

Tamil

Works on Rs. 800 feature phone

No app install. No smartphone.

2G / EDGE compatible.

Voice-first - no literacy required.

Rythu Seva Kendra Integration

We're not building a new system. We're amplifying what's already there.

The RSK Network

3,643 Rythu Seva Kendras across Telangana + AP.
Each has a trained officer.
Most underutilised - farmers don't know what they offer.

KisanAlert slots in:

- RSK officer opens dashboard on any laptop
- Flagged cases appear with photo + diagnosis
- Officer makes one call. Farmer gets expert help.
- Officer pushes district-wide alerts via dashboard

Scale: 0 to district-wide in days, not months.
No new hardware. No new centers.

3,643

RSK centers. Infrastructure already exists.

< 2 hours

Target SLA from flagged case
to officer field visit call.

From One District to the Country

Live Pilot

Month 1

Chittoor district, AP
~50 farmers via RSK officers
Manual onboarding

Early Access

Month 3

10 districts, Telangana + AP
500 farmers
RSK officers trained on dashboard

State Rollout

Month 6

Rythu Bandhu DB integration
Telangana: 2 lakh farmers reachable via
SMS

National

Year 2

PM-KISAN registry (10 Cr farmers)
Multi-state, multi-language
ISRO Bhuvan soil expansion

What makes this scalable:

- Cloud Run autoscales - a district is just a lat/lon lookup
- Adding a language = Gemini prompt change, not an engineering sprint
- RSK network already covers 3,643 centers nationally

Demo Walkthrough

Live at: <https://kisan-alert-pi.vercel.app>

Scenario 1

Crop Recommendation

Farmer (Warangal, Telugu): "ee yeda em panta veyali?"

KisanAlert checks: soil pH 6.4, GW 8m, kharif forecast

Response: Cotton #1 (88%), Soybean #2 (74%), Maize #3 (61%)

"Why?" panel shows: Soil pH, Moisture, NDVI trend, Rain 30d

Scenario 2

Dry Spell Alert

System detects: Open-Meteo shows 2.1mm / 7 days for Guntur

Auto-SMS: "agle 7 dinon mein barish nahin hogi. abhi sinchai karen."

Alert appears in RSK officer dashboard with severity tag
Officers can push district-wide from dashboard in one click

Scenario 3

Disease Diagnosis

Farmer Ramaiah sends groundnut leaf photo (Chittoor)

Gemini Vision: Leaf Spot (*Cercospora arachidicola*), 91% confidence

"Why 91%?" - matched 40,000+ ICAR labeled images

Treatment + RSK Chittoor Block 3 auto-assigned, SLA: 2h

Why This Will Actually Work

Working WITH government, not around it

Rythu Seva Kendras are funded, staffed, already trusted.
We give their officers better tools and a direct channel
to farmers who need urgent help.

KisanAlert earns adoption by making RSK officers
more effective - not by trying to replace them.

The AI cost structure works for this population

Per interaction	~2,000 tokens	~Rs. 0.013
1 lakh / month	full advisory	Rs. 1,260
District pilot	Cloud Run + Gemini + TTS	< Rs. 5,000/mo

Fundable from existing extension program budgets.

No new government spend required.

What we are NOT doing:

- Not replacing RSK officers. Supporting them.
- Not requiring smartphones. 2G SMS works.
- Not asking farmers to change behaviour. They already call for advice. We give them a number.

Technical Stack

All Google Cloud services. Containerised. Deploys in 10 minutes.

Gemini 1.5 Flash

NLU, crop recommendations,
multilingual conversation

Gemini Vision

Crop disease diagnosis
from farmer photos

Cloud STT / TTS

Voice input and output
for IVR calls

Cloud Run

Stateless, auto-scaling
containerised backend

FastAPI / Python

REST API backend
OpenAPI documented

SoilGrids + ISRO

Soil data at 250m resolution
National coverage

Open-Meteo + IMD

District-level 7-day forecast
+ historical rainfall norms

Vercel (frontend)

Static dashboard deployed
kisan-alert-pi.vercel.app

KisanAlert

The goal was never a demo.

*It was something an agricultural officer could open on
Monday morning and actually use.*

Live demo: <https://kisan-alert-pi.vercel.app>

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Gemini Powered

Multilingual

Voice First

RSK Ready